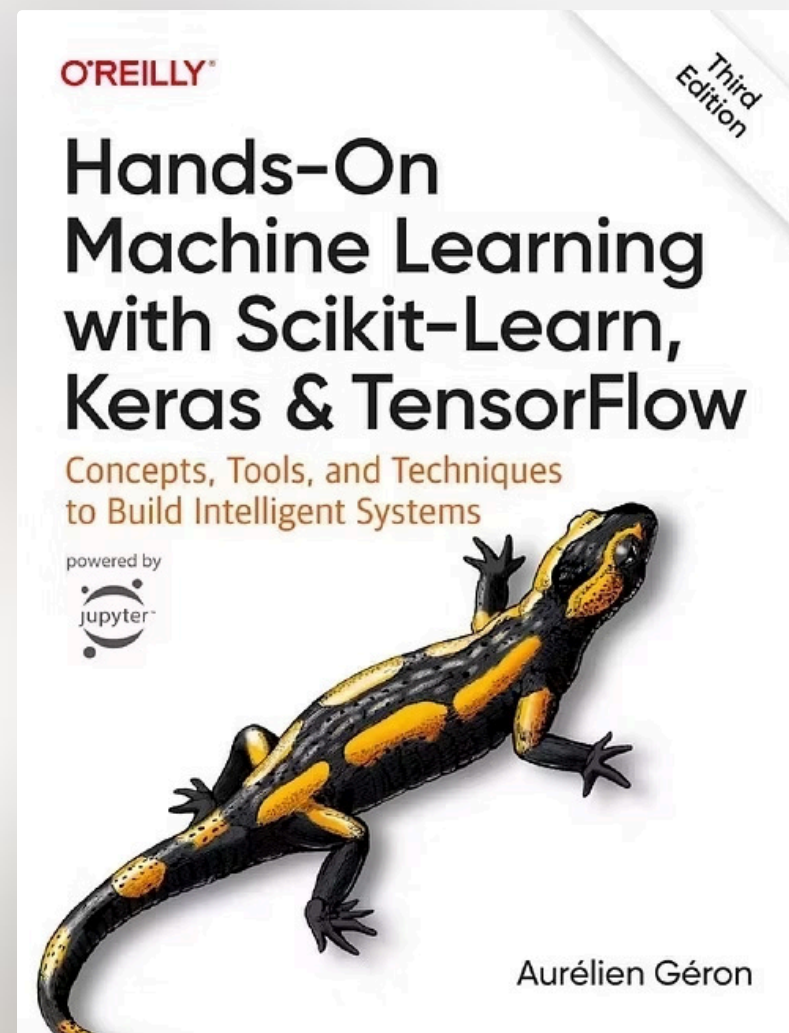


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/08/2025



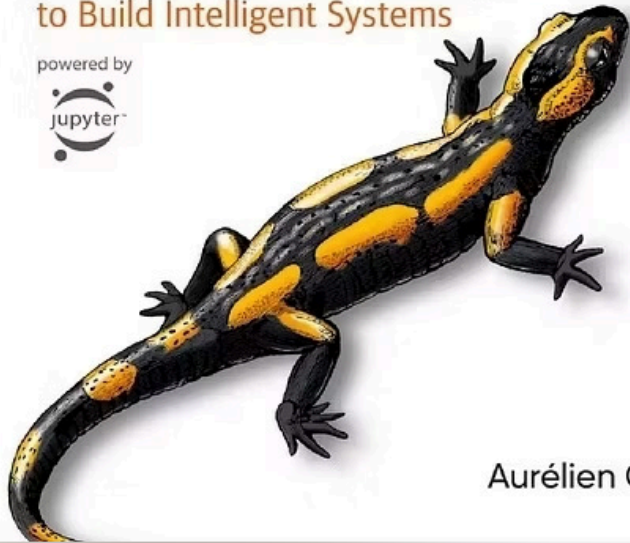
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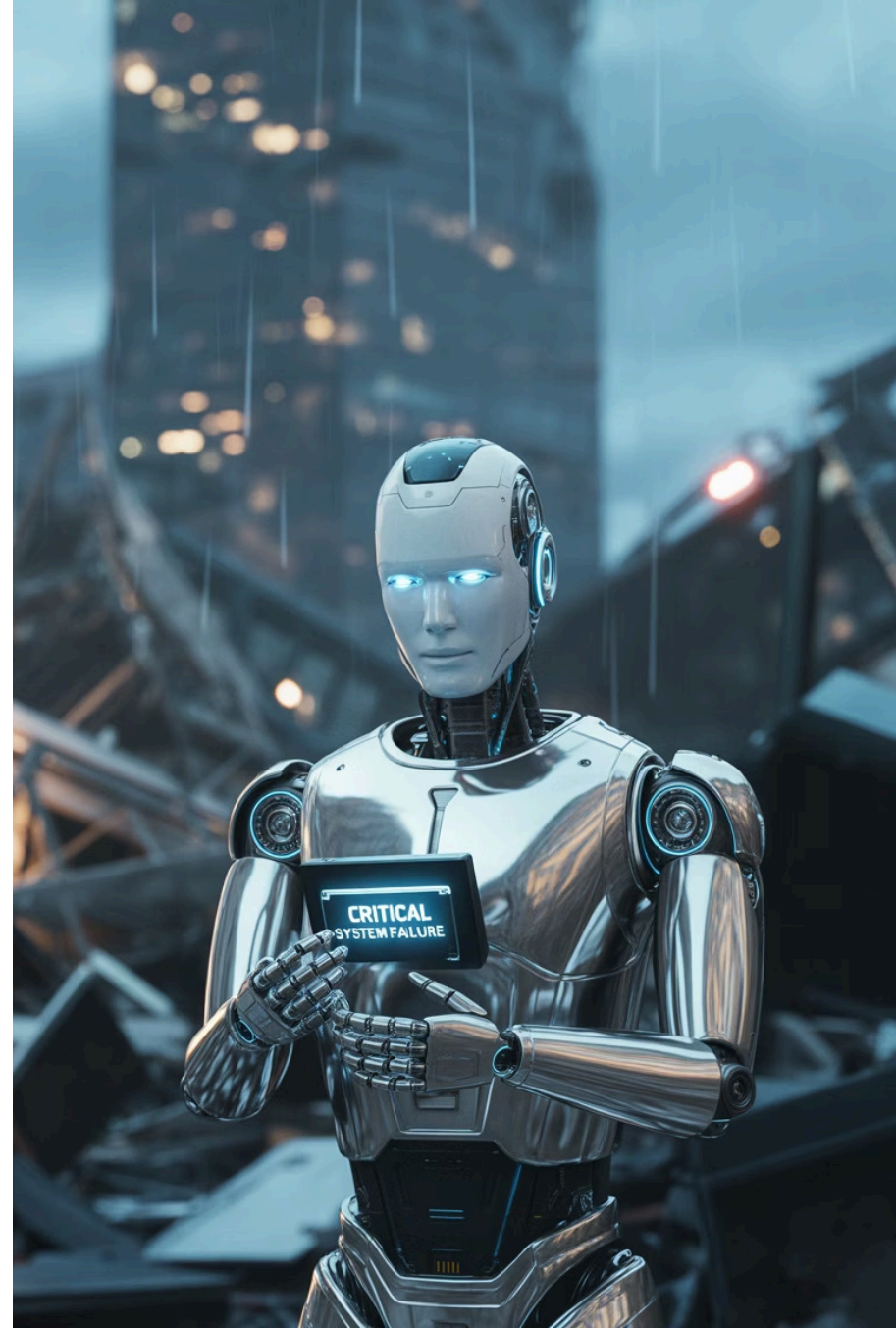
Aurélien Géron

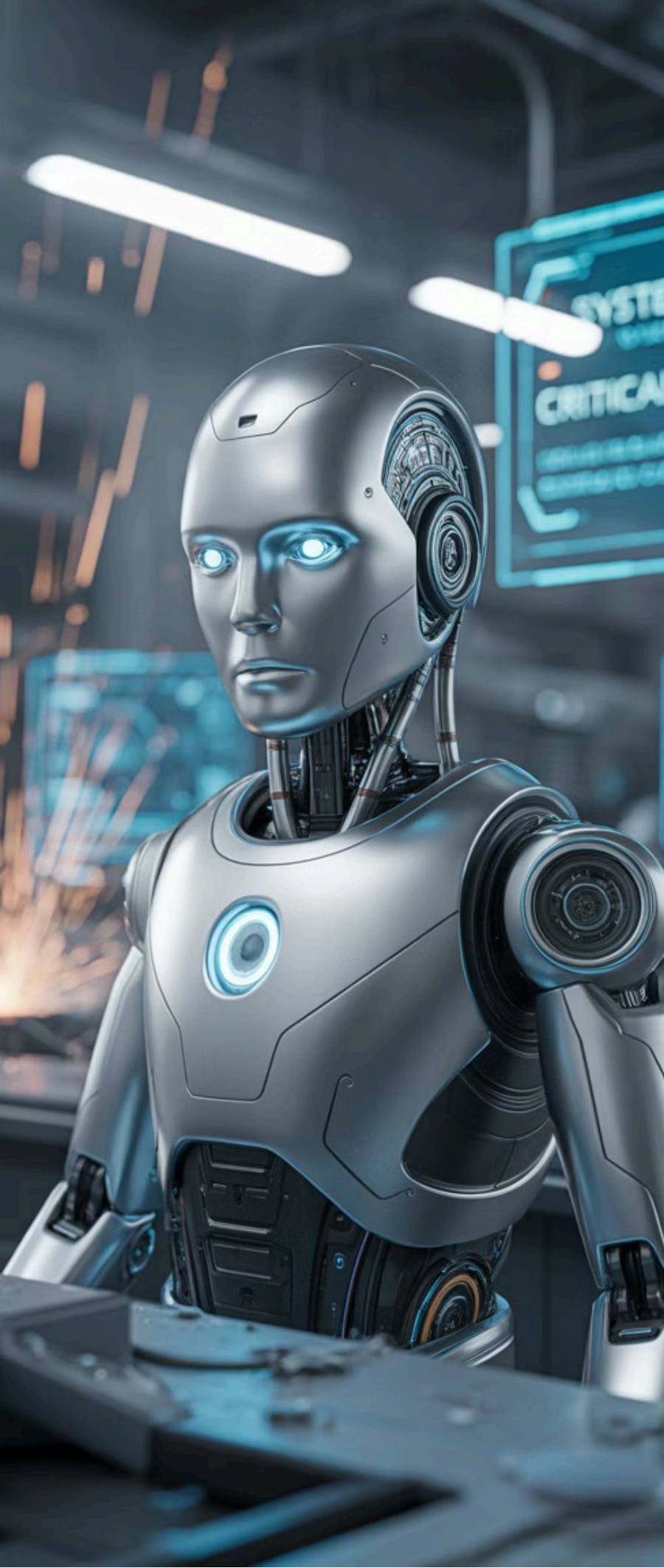
Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. **Testing and Validating**





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1:** Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - **Classification criteria #2:** Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3:** Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. **Testing and Validating**
 - **Hyperparameter Tuning and Model Selection**
 - Data Mismatch

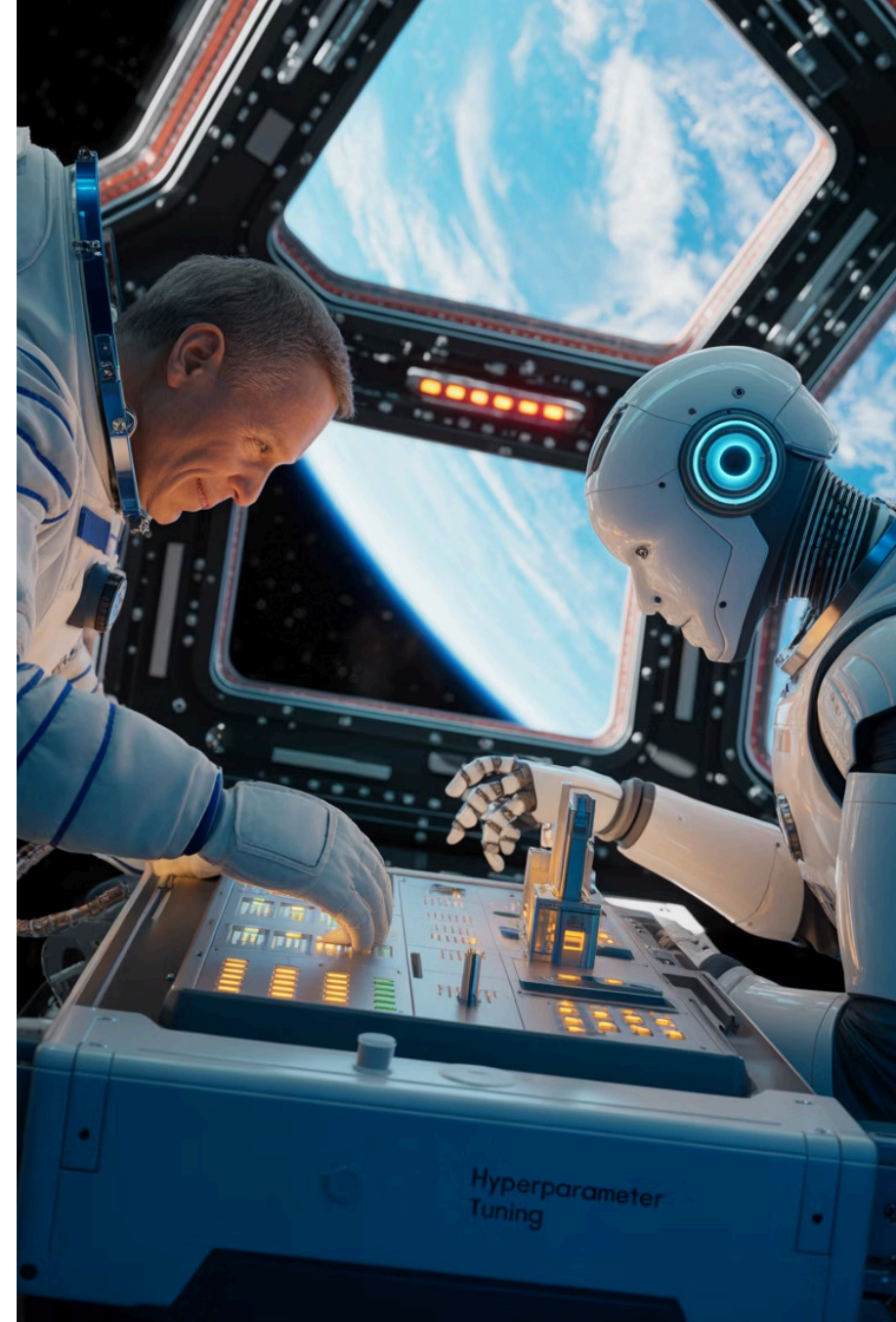
Testing and Validating

Hyperparameter Tuning and Model Selection, Data Mismatch

Hyperparameter Tuning and Model Selection: Overview

Problem:

- Deciding between **different types of models** (e.g., linear model vs. polynomial model).
- How to choose the **best model** and **tune its hyperparameters** for optimal performance.



Evaluating and Comparing Models

Step 1: Model Comparison:

Train multiple models (e.g., linear and polynomial) and compare their **generalization** using a **test set**.

Step 2: Hyperparameter Tuning:

After selecting the best model, **tune** its **hyperparameters** (e.g., **regularization parameter**) to **avoid overfitting**.

Common Approach: Train models with different hyperparameter values and **select** the one that **performs best on the test set**.



The Overfitting Pitfall

Problem:

If you **tune** the model's **hyperparameters** based on the **test set** performance, you **risk overfitting** to the **test set**.

Example:

A model may show **low generalization error** on the **test set** during **hyperparameter tuning**, but perform **poorly** in **production**.

Why? The model was **adapted too closely** to the **test set**, leading to **poor performance** on **new data**.



Solution: Holdout Validation

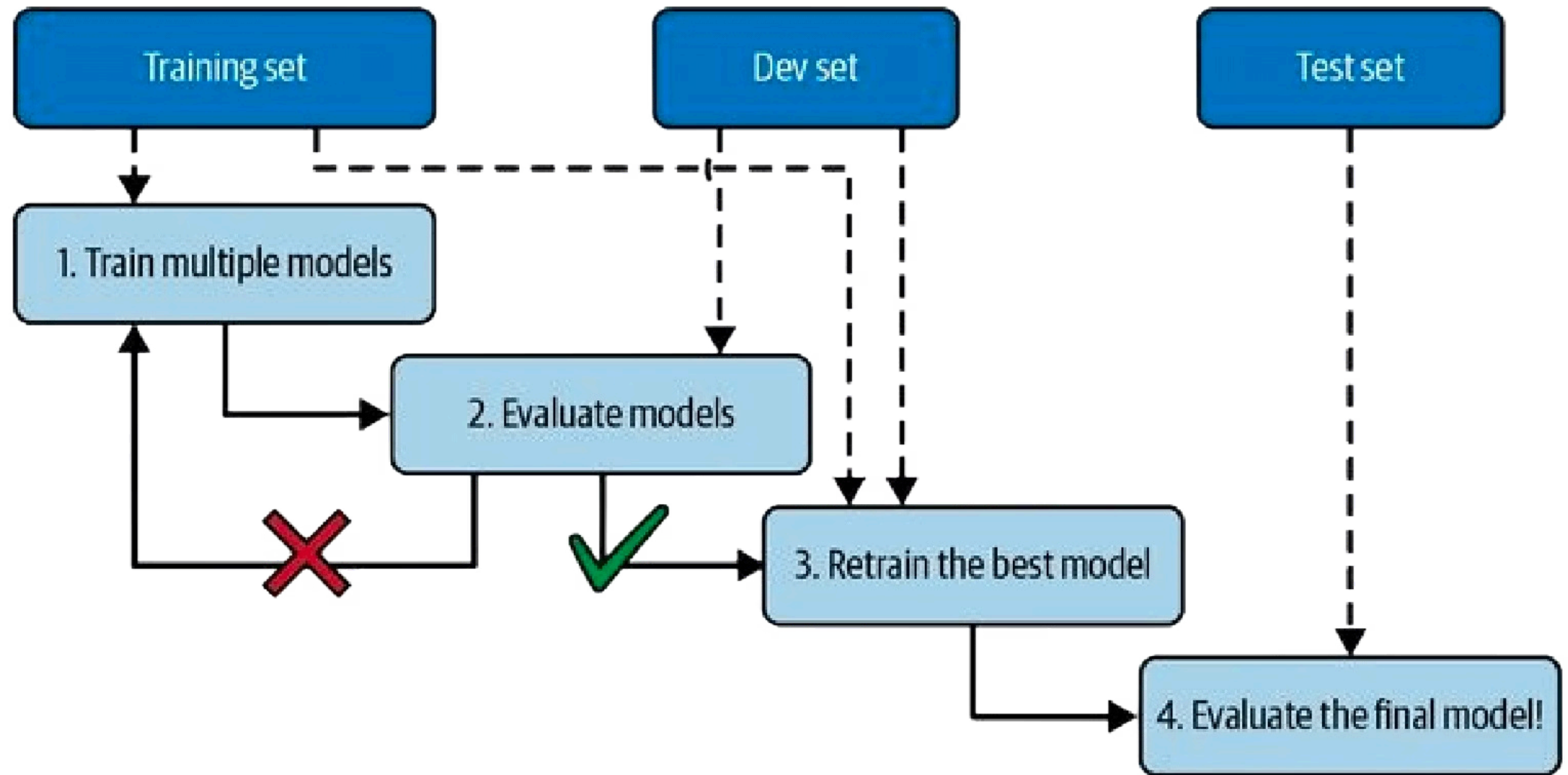


Figure 1-25. Model selection using holdout validation

Solution: Holdout Validation

What is Holdout Validation?

Holdout validation involves **splitting** the **data** into a **training set** and a **validation set**.

Process:



Train multiple models with different **hyperparameters** on the training set (**minus the validation set**).



Select the model that performs best on the **validation set**.



After selecting the best model, train it on the **full training set** (including the **validation set**).



Finally, evaluate the model's performance on the **test set** to estimate **generalization error**.

Benefit:

Helps **avoid overfitting** to the **test set** and ensures the **model generalizes** well to **new data**.

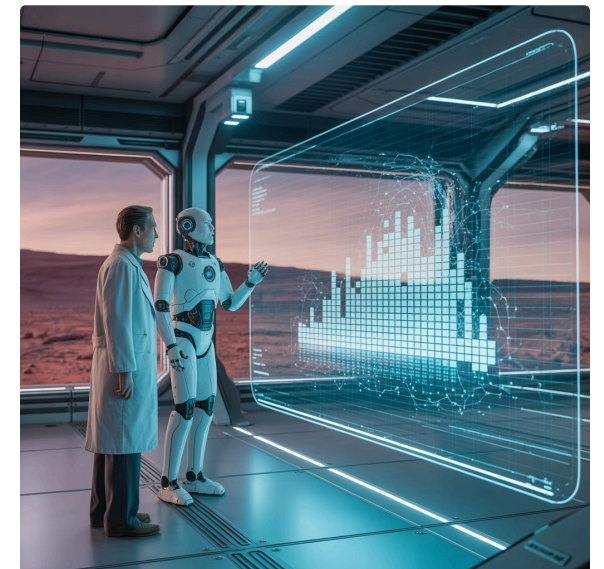
Challenges with Holdout Validation

Imprecision in Evaluations:

Small validation sets can lead to imprecise evaluations, causing the selection of a **suboptimal model**.

Problem with Large Validation Sets:

Large validation sets reduce the size of the **training set**, leading to less robust model training.



Solution: Cross-Validation

What is Cross-Validation?

Repeated cross-validation solves these problems by using **multiple smaller validation sets**.



Drawback:

Training time is multiplied by the number of validation sets, making the process computationally expensive.

Key Takeaways

Hyperparameter Tuning and Model Selection:

Tuning hyperparameters is crucial to optimize model performance but must be done without overfitting to the test set.

Holdout Validation:

A simple yet effective method for model selection and hyperparameter tuning using a validation set.

Cross-Validation:

Use cross-validation to improve evaluation accuracy by averaging results from multiple validation sets.



Next: Data Mismatch

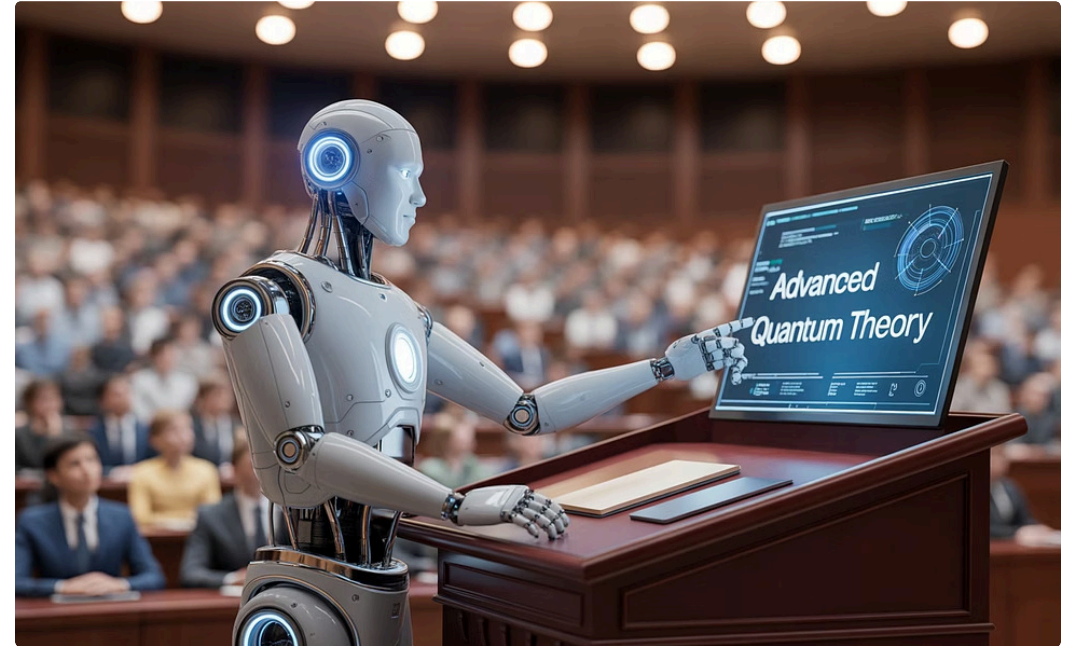


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts